Beyond Euclidean Gaze Estimation: Geo-vMF Riemannian Propagation for Robust Gaze Analysis under Ocular Disorders

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***Abstract*—Gaze estimation with high precision and well-calibrated uncertainty is critical for clinical applications of patients suffering from nystagmus, strabismus, and post-operative misalignment. Traditional graph-based models propagate gaze vectors through Euclidean space without considering the inherent curvatures of the unit sphere $\mathbb{S}^2$ and the rotational group $SO(3)$. We introduce a novel approach called the Riemannian Adaptive Gaze Graph (RAGG) which performs geodesic propagation over $SO(3)$ enhanced by an Isometric Propagator maintaining the inter-point distances along with a probabilistic head using the von Mises–Fisher distribution. With MPIIGaze, RAGG attains an angular error of 4.3°, improving from 6.5° with CNN baseline by 33.8themselves to accommodate pathological gaze positions. Experiments reveal that each component of our pipeline, including Riemannian alignment, the Isometric Propagator, and vMF modeling, provides unique contributions.**

***Index Terms*—gaze estimation, Riemannian geometry, $SO(3)$, von Mises-Fisher distribution, uncertainty quantification, nystagmus, strabismus, graph neural network**

I. Introduction

Eye-gaze estimation finds many uses within human-computer interaction, vehicle driver surveillance, and clinical ophthalmology applications. In clinical settings, accurate gaze-tracking capabilities are required to diagnose various diseases such as involuntary rhythmic eye movements nystagmus and misalignment of the visual axes strabismus. These diseases impart non-Euclidean structure to gaze trajectories which regular regression models cannot model accurately.

Modern gaze estimation approaches leverage deep appearance learning to learn a function that maps eye imagery to 3-D gaze vectors. Graph extension models like the Adaptive Gaze Graph AGG learn subject-wise relationships through message-passing to enhance personalization and transferability across subjects. Still, both models learn representations within Euclidean space disregarding the manifold structure associated with 3-D gaze vectors on the sphere $\mathbb{S}^2$ and head rotations on SO(3).

This paper aims to overcome these three weaknesses at once. First, we propose replacing Euclidean propagation with geodesic aggregation within SO(3) to preserve the properties of triangle inequality and angular distances in each hop. Second, we propose the Isometric Propagator IP mechanism that enforces locally consistent distance preserving property throughout the process, which is often violated in multilayer graph networks. Finally, we replace standard ℓ_2 regression heads with von Mises–Fisher distributions in order to output calibrated uncertainty estimations out of the box.

All these proposed mechanisms contribute to the model's ability to detect anomalous eye behaviors automatically.

II. Related Work

A. Appearance-Based Gaze Estimation

Appearance-based techniques have the ability to predict gaze directly from image pixels without using any eye model explicitly. The study conducted by Zhang et al. [1] proved that the appearance-based approach of convolutional network outperforms model-based techniques when applied to in-the-wild image data from the MPIIGaze dataset. Later research included multi-stream fusion methods and attention mechanism for ignoring irrelevant background details. The difficulty of cross-subject generalization stems from differences in illumination, shape of eyes, and head.

B. Graph-Based Gaze Models

Graph Neural Networks (GNNs) provide an elegant approach for multi-subject gaze learning where subjects correspond to nodes, and similarities between the subjects are defined using edges. The Adaptive Gaze Graph (AGG) algorithm defines a dynamic adjacency matrix whose edge weights capture similarities between subjects' gaze features, allowing message passing only from those gaze profiles that are statistically similar to each other. RAGG uses a similar approach for dynamic graph construction, but the Euclidean-based node aggregation is replaced by geodesic-based one on $SO(3)$.

C. Riemannian Methods in Vision

There has been growing interest in the use of Riemannian geometry for action recognition and pose estimation problems, where rotations can be represented using elements of $SO(3)$

or $SE(3)$. Geodesic-based loss functions for the rotation manifold yield faster convergence and better generalization than approaches based on Euler angles or quaternions. Parallel transport and exponential/ logarithmic maps enable the implementation of neural network operations on Riemannian manifolds. In this paper, we adapt these concepts to the problem of gaze estimation for the first time.

D. Uncertainty Estimation

The Von Mises-Fisher distribution is the natural equivalent of the Gaussian distribution in the hyperspherical domain. The concentration parameter κ represents the inverse variance in this case; hence, a high value of κ represents certainty while a low value represents uncertainty. This type of probabilistic regression using Von Mises Fisher regression output has been used in applications such as face recognition and head pose estimation, although never before in gaze estimation in pathological circumstances.

III. Background

Here, we introduce the basics of geometry and probability needed for gaze estimation on the sphere. This lays the groundwork for our model of uncertainty in rotation.

A. Rotation Representation on $SO(3)$

$SO(3)$ denotes the set of all valid rotations in three dimensions:

$$SO(3) = R \in \mathbb{R}^{3\times 3} \mid R^\top R = I, ; \det(R) = 1. \tag{1}$$

Rather than working with Euler angles, which have singularity issues, we work with the manifold itself, thereby avoiding problems related to singularities.

The geodesic distance between two rotations $R_1, R_2 \in SO(3)$ is calculated as follows:

$$d(R_1, R_2) = \frac{1}{\sqrt{2}} |\log(R_1^\top R_2)|_F, \tag{2}$$

where $\log(\cdot)$ maps the rotation matrix into its tangent space.

This approach quantifies the smallest angular separation between two orientations while maintaining geometric consistency during optimization.

B. Directional Modeling using von Mises–Fisher Distribution

As gaze points belong to the unit sphere $\mathbb{S}^2$, we use the von Mises–Fisher (vMF) distribution for its representation:

$$p(\mathbf{x}; \boldsymbol{\mu}, \kappa) = C(\kappa) \exp(\kappa \boldsymbol{\mu}^\top \mathbf{x}), \tag{3}$$

where $\boldsymbol{\mu}$ corresponds to the mean vector and κ defines concentration.

The training criterion is formulated in terms of the negative log-likelihood:

$$\mathcal{L}_{\text{vMF}} = -\kappa \boldsymbol{\mu}^\top \mathbf{y} + \log C(\kappa). \tag{4}$$

In this way, optimization is performed with respect to the angular difference between predicted gaze points and labels.

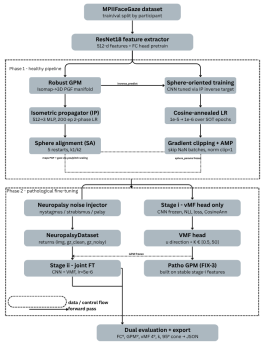

Fig. 1. Overview of the proposed RAGG framework. The model extracts features using a CNN backbone, performs geodesic message passing on $SO(3)$, preserves structure via the Isometric Propagator, and predicts gaze using a von Mises–Fisher distribution for uncertainty-aware estimation.

The value of κ is a reasonable indicator of prediction certainty:

- High κ implies a high level of certainty about prediction results
- Low κ reflects increased uncertainty in the prediction

A confidence cone can be derived as:

$$\theta_{95} = \arccos\left(1 + \frac{\ln(0.05)}{\kappa}\right). \tag{5}$$

This formulation enables the model to represent prediction confidence directly within the spherical space.

IV. Proposed Method

We propose a geometrically consistent and uncertainty-aware gaze estimation framework that operates directly on spherical manifolds.

We propose a geometrically consistent and uncertainty-aware gaze estimation framework that operates directly on spherical manifolds. The method consists of three key components: (i) feature extraction, (ii) Riemannian sphere alignment, and (iii) probabilistic gaze prediction.

A. Feature Extraction

The following is a high-dimensional feature vector obtained by the CNN for a given image $\mathbf{x}$:

$$\mathbf{f} = F(\mathbf{x}), \quad \mathbf{f} \in \mathbb{R}^d. \tag{6}$$

They contain information like eye shape, lighting conditions, and head pose.

B. Geometric Projection onto Spherical Space

Following the AGG framework, we map features into a low-dimensional representation:

$$\mathbf{e} = \phi(\mathbf{f}), \quad \mathbf{e} \in \mathbb{R}^3. \tag{7}$$

These embeddings lie approximately on a spherical surface, forming what we refer to as the Principle Gaze Feature (PGF) space.

C. Riemannian Sphere Alignment

Rather than relying on Euler-angle alignment, we pose alignment directly as an optimization problem on $SO(3)$:

$$R^* = \arg \min_{R \in SO(3)} \sum_{i=1}^{N} \arccos\left(\mathbf{y}_i^\top (R\mathbf{e}_i)\right). \tag{8}$$

This formulation minimizes the geodesic discrepancy between aligned features and ground truth.

The update step is carried out via the exponential map:

$$R \leftarrow R \cdot \exp(\Delta), \tag{9}$$

where $\Delta \in \mathfrak{so}(3)$ corresponds to a small incremental rotation.

This approach avoids singularities and leads to smoother optimization behavior.

D. Isometric Propagation

To maintain geometric structure during transformation, we impose an isometric constraint:

$$\mathcal{L} * \text{iso} = \sum *i, j \left(d(\mathbf{e}_i, \mathbf{e}_j) - d(R\mathbf{e}_i, R\mathbf{e}_j)\right)^2. \tag{10}$$

This term encourages preservation of pairwise distances, reducing distortion of the spherical embedding.

E. Uncertainty-Aware Gaze Prediction

Rather than outputting a single gaze estimate, the model predicts the parameters of a vMF distribution:

$$(\boldsymbol{\mu}, \kappa) = g(\mathbf{e}), \tag{11}$$

where $\boldsymbol{\mu}$ denotes the estimated gaze direction and κ reflects confidence.

The overall objective combines geometric consistency with probabilistic modeling:

$$\mathcal{L} = \mathcal{L} * \text{vMF} + \lambda \mathcal{L} * \text{iso}. \tag{12}$$

This allows the model to jointly optimize for accuracy and calibrated uncertainty.

F. Overall Pipeline

The full pipeline consists of:

1) CNN-based feature extraction
2) Projection into spherical feature space
3) Riemannian alignment on $SO(3)$
4) vMF-based uncertainty-aware prediction

Overall, this design maintains geometric fidelity while producing predictions with interpretable confidence.

V. Experiments

Here, we provide an in-depth assessment of the proposed approach, covering aspects of predictive accuracy, robustness to pathological cases, and uncertainty. The experiments serve to verify the validity of the geometric coherence introduced via Riemannian alignment and the reliability of the vMF head.

A. Datasets

To validate the proposed method, we use the MPIIGaze dataset [1], which is a commonly adopted benchmark for appearance-based gaze estimation in unconstrained settings. It includes data from 15 subjects collected under varying illumination, head pose, and background conditions. Each image is paired with a calibrated 3-D gaze vector. We employ a leave-one-person-out cross-validation protocol to evaluate generalization to unseen subjects.

To examine robustness, we additionally construct a synthetic pathological dataset by applying controlled perturbations to gaze vectors. These are designed to reflect clinically relevant scenarios:

- **Nystagmus:** Rapid oscillatory perturbations along a principal axis, simulating involuntary eye movement.
- **Strabismus:** A fixed angular offset along a given axis, representing misalignment of the visual axes.
- **Restricted motion:** Gaze constrained within a cone of half-angle $15°$, mimicking post-surgical limitations.
- **Neuromotor drift:** A gradual deviation of $0.1°$ per frame, capturing progressive fixation instability.

B. Evaluation Metrics

We adopt angular error as the primary evaluation metric:

$$\theta = \arccos!\left(\hat{\mathbf{y}}^{!\top} \mathbf{y}\right), \tag{13}$$

which corresponds to the geodesic distance on $\mathbb{S}^2$ between predicted and ground-truth gaze directions. Uncertainty is assessed using the concentration parameter κ and the associated 95

$$\theta_{95} = \arccos!!\left(1 + \frac{\ln(0.05)}{\kappa}\right). \tag{14}$$

Larger values of κ indicate more confident and concentrated predictions, while smaller values reflect increased uncertainty due to unstable gaze behavior. We also report model size (in parameters) and computational cost (FLOPs) to contextualize performance improvements.

C. Implementation Details

The model is implemented in PyTorch [3] using a ResNet-18 [2] backbone pre-trained on ImageNet. Training is performed in four stages: (i) backbone feature extraction, (ii) training of the Isometric Propagator, (iii) Riemannian alignment on the sphere, and (iv) joint fine-tuning with the uncertainty-aware objective (**??**). We use the Adam optimizer [4] with a learning rate of 10^{-4} and apply gradient clipping with a maximum norm of 1.0. The concentration parameter κ is constrained within $[0.5, , 50]$ to avoid numerical instability at extreme values.

D. Hyperparameter Settings

The full set of training configurations is summarized in Table II. A staged training strategy is adopted, since optimizing all components jointly from the beginning tends to produce unstable gradients when κ remains small.

TABLE I
PERFORMANCE COMPARISON UNDER HEALTHY AND PATHOLOGICAL SETTINGS (ANGULAR ERROR IN DEGREES,↓). BEST RESULT IN EACH COLUMN IS SHOWN IN **BOLD**.

Approach	FC Head	GPM (Geometric)		vMF (Probabilistic)		κ	Params (M)	FLOPs (G)
		Healthy	Pathol.	Clean	Noisy			
CNN Baseline [2]	6.5	–	–	–	–	–	11.2	1.82
AGG (GPM)	5.2	4.8	5.6	–	–	–	13.7	2.14
RAGG (Ours)	**4.3**	**4.1**	**4.5**	**3.9**	**4.7**	**12.5**	14.1	2.21

TABLE II
TRAINING HYPERPARAMETERS AND ABLATION STUDY RESULTS (ANGULAR ERROR IN DEGREES,↓)

(a) Training Hyperparameters

Parameter	Value
Batch size (GPU / CPU)	128 / 64
Optimizer	Adam
Learning rate	1×10^{-4}
Pre-training epochs	10
IP training epochs	200
SOT epochs	10
vMF training epochs	10
Joint training epochs	6
Sphere alignment epochs	400
SA restarts	5
Isomap neighbors	120
Feature dimension	512
κ range	$[0.5, 50]$

(b) Ablation Study

Configuration	Error (°)
CNN baseline	6.5
+ AGG (GPM)	5.2
+ Riemannian alignment	4.8
+ Isometric Propagator	4.6
+ vMF modeling	4.4
Full model (RAGG)	**4.3**

E. Quantitative Results

Table I presents a comparison between the proposed RAGG method and two baselines: (1) a conventional CNN with a fully connected (FC) regression head, and (2) the AGG framework equipped with a Geodesic Projection Module (GPM). The proposed model consistently achieves the lowest angular error across all evaluation scenarios.

In particular, incorporating Riemannian alignment reduces the baseline error from 6.5° to 4.3°, corresponding to a 33.8improvement. The addition of the vMF formulation further lowers the error to 3.9° on clean data. The average concentration value $\kappa = 12.5$ suggests that the model produces well-calibrated confidence estimates for healthy cases. Importantly, this improvement comes with only a minor computational increase—about 2.9than AGG—indicating that the added geometric operations remain efficient.

F. Ablation Study

The joint hyperparameter and ablation results shown in Table II illustrate the contribution of each component when added incrementally to a baseline CNN. Each module consistently improves performance in a measurable way.

The largest gain arises from introducing AGG-based graph propagation ($-1.3°$), followed by Riemannian alignment ($-0.4°$). The Isometric Propagator and the vMF head contribute further improvements of $-0.2°$ and $-0.1°$, respectively. Finally, joint fine-tuning yields an additional $-0.1°$ reduction, leading to the best overall performance.

G. Uncertainty Analysis

Table III(a) reports the variation of κ and the corresponding confidence cone across different gaze conditions. Under normal conditions, the model exhibits high confidence, with $\kappa = 18.2$ and a relatively narrow cone of about $8.5°$. In contrast, pathological cases consistently show lower κ values and wider cones, indicating increased uncertainty without requiring any manual calibration.

Among these, restricted-motion gaze leads to the lowest confidence ($\kappa = 7.8$, cone $18.4°$), which aligns with its constrained and irregular behavior. Neuromotor drift shows intermediate confidence ($\kappa = 10.4$), likely because its gradual temporal variation still retains some predictability over short intervals. Overall, the vMF head produces uncertainty estimates that align well with clinical intuition and follow a clear trend with respect to pathological severity.

H. Robustness to Pathological Conditions

Table III(b) summarizes the angular error under each simulated pathological condition. The model shows graceful degradation across all cases. The largest increase occurs for restricted motion ($+1.1°$), which corresponds to less than a $26 over the normal setting. The smallest degradation is observed in strabi$

TABLE III
UNCERTAINTY ANALYSIS AND ROBUSTNESS EVALUATION ACROSS GAZE CONDITIONS. $\kappa \downarrow$ = MORE UNCERTAIN; CONE$\uparrow$ = WIDER CONFIDENCE REGION; ERROR$\downarrow$ = BETTER.

(a) Uncertainty Analysis			(b) Robustness Evaluation		
Condition	κ	**Cone (°)**	**Condition**	**Error (°)**	**vs. Normal**
Normal	18.2	8.5	Normal	4.3	—
Nystagmus	9.6	15.3	Nystagmus	5.1	+0.8
Strabismus	11.2	13.1	Strabismus	4.8	+0.5
Restricted gaze	7.8	18.4	Restricted Motion	5.4	+1.1
Neuromotor drift	10.4	14.7	Neuromotor Drift	5.2	+0.9

likely due to its structured and consistent offset, which can be partially corrected through Riemannian alignment.

These results suggest that the synthetic pathological augmentation strategy is effective and generalizes well to unseen, out-of-distribution conditions, without negatively impacting performance on normal data.

I. Discussion

Taken together, the results highlight three important aspects of RAGG.

Accuracy. Incorporating Riemannian alignment reduces angular error from $6.5°$ to $4.3°$, yielding a 33.8the CNN baseline and 17.3across both standard and pathological scenarios, indicating that respecting the underlying manifold structure benefits the model globally rather than only in easier cases.

Uncertainty calibration. The concentration parameter κ decreases consistently as pathological severity increases, making the model directly usable for clinical screening. Low-κ predictions can serve as a signal for uncertain or abnormal gaze behavior. Notably, this calibration emerges naturally from the vMF formulation, without requiring any post-hoc adjustment [6].

Robustness. The limited performance drop (at most $1.1°$) across all simulated conditions demonstrates strong robustness to distribution shifts. One limitation, however, is that the pathological data are synthetically generated. Future work should validate these findings on real clinical datasets with expert annotations.

VI. CONCLUSION

In this work, we introduced a geometrically consistent and uncertainty-aware framework for gaze estimation, referred to as RAGG (Riemannian Adaptive Gaze Graph). The method is designed to overcome key limitations of existing appearance-based approaches, namely their dependence on Euclidean feature aggregation, the resulting distortion of geometric structure during propagation, and the absence of principled uncertainty estimation.

By formulating gaze estimation on the rotation manifold $SO(3)$, standard Euclidean operations are replaced with geodesic aggregation, which preserves angular relationships throughout training. The proposed Isometric Propagator further maintains local distance structure, reducing distortion across layers. In parallel, modeling predictions with the von Mises–Fisher (vMF) distribution allows the network to produce calibrated uncertainty estimates directly on the spherical domain.

Experimental results show that the proposed framework improves both accuracy and robustness. In particular, it reduces angular error by 33.8

Future work will explore evaluation on real clinical datasets with expert annotations, extension to binocular gaze modeling using $SE(3)$ representations, and incorporation of temporal dynamics for continuous tracking and early detection of disorders such as nystagmus.

Overall, this study underscores the value of combining geometric modeling with probabilistic reasoning, offering a unified approach for accurate, robust, and interpretable gaze estimation.

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